

RAM CHARAN VANGA

RA2511027010164

Sub ject : DSA

Unit 3, S1, SLO1 - Stacks Introduction

1. A stack is a ____ data structure.

Answer: linear (LIFO - Last In First Out)

2. The two main operations of a stack are ____ and ____.

Answer: push and pop

3. The operation that adds an element to a stack is called ____.

Answer: push

4. The operation that removes the top element from the stack is called ____.

Answer: pop

5. The operation that returns the top element without removing it is called ____.

Answer: peek (or top)

6. If a stack is full and a push operation is attempted, it is called ____.

Answer: stack overflow

7. If a stack is empty and a pop operation is attempted, it is called ____.

Answer: stack underflow

8. Stacks are usually implemented using ____ or ____.

Answer: arrays or linked lists

9. In array implementation of stack, the top element is accessed using the ____ index.

Answer: TOP

10. In function calls, the system uses a stack to store ____.

Answer: activation records (return addresses, local variables, and parameters)

11. The initial value of top in an empty stack array implementation is usually ____.

Answer: -1

12. Stack is used in ____ notation to evaluate expressions without parentheses.

Answer: postfix (Reverse Polish) / prefix

13. A stack is also used in ____ traversal of trees.

Answer: depth-first (preorder, inorder, postorder)

14. The time complexity of push and pop operations in a stack is ____.

Answer: $O(1)$ (constant time)

15. Reversing a string using stack works by ____ each character and then ____ them.

Answer: pushing each character and then popping them